Module 1: Introduction to NLP

1.1 Explain the applications of Natural Language processing. [5M] (Frequency: 2)

1.2 Illustrate the concept of tokenization and stemming in Natural Language processing. [5M]

1.3 Explain the challenges of Natural Language processing. [5M] (Frequency: 2)

1.4 Explain the various stages of Natural Language processing. [5M] (Frequency: 3)

1.5 Explain Different stage involved in NLP process with suitable example. [10]

1.6 Discuss the challenges in various stages of natural language processing. [10]

1.7 Explain following Syntactic and Semantic Constraints on Co reference 1) Number Agreement 2) Person & Case Agreement [5M]

1.8 Explain different stages of NLP. Also explain generic NLP system. [10]

1.9 Explain the ambiguities associated at each level with example for Natural Language processing. [10]

1.10 Differentiate between Syntactic ambiguity and Lexical Ambiguity. [5M] (Frequency: 2)

1.11 What is the rule-based and stochastic part of speech taggers? [5M]

1.12 What is hybrid POS tagging? [5M]

1.13 Explain pre-processing steps generally used in NLP. [5M]

1.14 Explain the preprocessing operations in natural language processing [10]

Module 2: Word Level Analysis

2.1 Compare Derivational & Inflectional morphology [5M] (Frequency: 2)

2.2 Explain inflectional and derivational morphology with an example [10]

2.3 What is the output of Morphological Analysis for Regular Verb, Irregular verb, Singular noun, Plural noun. [5M] (Frequency: 2)

2.4 Represent output of morphological analysis for Regular verb, Irregular verb, singular noun, plural noun Also Explain Role of FST in Morphological Parsing with an example [10]

2.5 Illustrate the working of Porter stemmer algorithm [10] (Frequency: 2)

2.6 Explain Porter Stemmer algorithm with rules [10] (Frequency: 4)

2.7 Explain Porter Stemming algorithm in detail. [10] (Frequency: 3)

2.8 Design FST for regular and plural nouns. [10]

2.9 Explain FSA for nouns and verbs. Also Design a Finite State Automata (FSA) for the words of English numbers 1-99. [10]

2.10 Explain the role of FSA in morphological analysis? [10]

2.11 Explain concepts of Bi-gram and n-gram with formula. [10]

2.12 Explain the N-gram language model and its application. [10]

2.13 Explain how N-gram model is used in spelling correction [5M]

2.14 Explain Good Turing Discounting? [5M] (Frequency: 2)

2.15 Explain perplexity of any language model. [5M]

2.16 For following corpus, apply Bi-gram model [10] Training Corpus: <s> I am Sam </s> <s> Sam I am </s> <s> Sam I like </s> <s> Sam I do like </s> <s> do I like Sam </s>

1. What is the most probable next word predicted by the model for the following word sequences?

(a) <s> Sam . . . (b) <s> Sam I do . . . (c) <s> Sam I am Sam . . . (d) <s> do I like . . .

2. Which of the following sentences is better, i.e., gets a higher probability with this model?

(e) <s> Sam I do I like </s> (f) <s> Sam I am </s> (g) <s> I do like Sam I am </s>

2.17 Consider the following corpus <s> I tell you to sleep and rest </s> <s> I would like to sleep for an hour </s> <s> Sleep helps one to relax </s> List all possible bigrams. Compute conditional probabilities and predict the next word for the word “to”. [10]

2.18 Define affixes. Explain the types of affixes. [5M]

2.19 Describe open class words and closed class words in English with examples. [5M]

2.20 Explain the significance of regular expression in NLP. [10]

2.21 What are the limitations of Hidden Markov Model (HMM) and MaxEnt Model for POS Tagging. [5M]

Module 3: Syntax Analysis

3.1 Discuss the challenges in part of speech tagging [5M] (Frequency: 2)

3.2 Explain the various challenges in POS tagging. [10]

3.3 Explain hidden markov model for POS based tagging. [10]

3.4 Explain Maximum Entropy Model for POS Tagging [10]

3.5 Demonstrate the concept of conditional random field in NLP [10]

3.6 Explain how Conditional Random Field (CRF) is used for sequence labeling. [10] (Frequency: 2)

3.7 Explain Shift Reduce Parser in NLP with example [10]

3.8 For a given grammar using CYK or CKY algorithm parse the statement “The man read this book”

Rules:

S → NP VP | Aux NP VP | VP

NP → Det NOM

NOM → Noun | Noun NOM

VP → Verb | Verb NP

Det → that | this | a | the

Noun → book | flight | meal | man

Verb → book | include | read

Aux → does

[10] (Frequency: 2)

3.9 Construct a parse tree for the following sentence using the given CFG rules: [10] The tall girl sings.

Rules: S → NP VP

NP → Det Adj N | Det N

VP → V | V NP

Det → “the”

Adj → “tail”

N → “girl”

V → “sings”

3.10 Explain the use of Probabilistic Context Free Grammar (PCFG) in natural language processing with example. [10]

3.11 Compare top-down and bottom-up approach of parsing with example. [10]

3.12 What are the limitations of Hidden Markov Model? [10]

3.13 Consider the following corpus and compute emission/transition probabilities for a bigram HMM and decode a given sentence using Viterbi. [10]

* Variant 1: Corpus: <s> a/DT dog/NN chases/V a/DT cat/NN </s> <s> the/DT dog/NN barks/V loudly/RB </s> <s> a/DT cat/NN runs/V fast/RB </s>. Decode: “The cat chases the dog.”

* Variant 2: Corpus: <s> the/DT students/NN pass/V the/DT test/NN<s> <s> the/DT students/NN wait/V for/P the/DT result/NN<s> <s> teachers/NN test/V students/NN<s>. Decode: “The students wait for the test”

* Variant 3 (Table-based):

[Table: <S> Martin Justin can watch Will <E> <S> Spot will watch Martin <E> <S> Will Justin spot Martin <E> <S> Martin will pat Spot <E>]

N: Noun [Martin, Justin, Will, Spot, Pat], M: Modal verb [can , will], V: Verb [watch, spot, pat]

Create Transition & Emission Matrices. Tag: “Justin will spot Will”. [10] (Frequency: 2)

Module 4: Semantic Analysis

4.1 Describe the semantic analysis in Natural Language processing. [5M]

4.2 Demonstrate lexical semantic analysis using an example [10]

4.3 What is word sense disambiguation? [5M] (Frequency: 2)

4.4 What is Word Sense Disambiguation (WSD)? Explain the dictionary based approach to Word Sense Disambiguation. [10] (Frequency: 2)

4.5 What do you mean by word sense disambiguation (WSD)? Discuss dictionary based approach for WSD. [10]

4.6 Why there is need of word sense disambiguation [10]

4.7 Explain Lesk algorithm for Word Sense Disambiguation. [10] (Frequency: 2)

4.8 Explain dictionary-based approach (Lesk algorithm) for word sense disambiguation (WSD) with suitable example. [10]

4.9 Explain Naive Bayes Supervised algorithm for Word sense Disambiguation [10]

4.10 Explain how the supervised learning approach can be applied for word sense disambiguation [10]

4.11 Explain Semi-supervised method (Yarowsky) Unsupervised (Hyperlex) [10] (Frequency: 2)

4.12 Explain Yarowsky bootstrapping approach of semi supervised learning [10]

4.13 Explain with suitable example the following relationships between word meanings: Hyponymy, Hypernymy, Meronymy, Holynymy [5M]

4.14 Explain with suitable example following relationships between word meanings. Homonymy, Polysemy, Synonymy, Antonymy [5M]

Module 5: Pragmatic & Discourse Processing

5.1 Explain reference resolution in detail [5M] (Frequency: 2)

5.2 What is reference resolution? [5M]

5.3 Explain Discourse reference resolution in detail. [10]

5.4 Illustrate the reference phenomena for solving the pronoun problem [10]

5.5 Explain Anaphora Resolution using Hobbs and Cantering Algorithm [10] (Frequency: 3)

5.6 How Anaphora Resolution is performed with Hobbs and Centering Algorithm [10]

5.7 Describe in detail Centering Algorithm for reference resolution. [10]

5.8 Explain Hobbs algorithm for pronoun resolution. [10]

5.9 Compare and contrast Hobbs' Algorithm and Centering Theory. [10]

5.10 What are five types of referring expressions? Explain with the help of example. [10]

5.11 Explain three types of referents that complicate the reference resolution problem. [5M]

Module 6: Applications of NLP

6.1 Explain Machine Translation Approaches used in NLP. [5M] (Frequency: 2)

6.2 Explain rule-based machine translation systems [5M] (Frequency: 2)

6.3 Explain statistical approach for machine translation. [5M]

6.4 Demonstrate the working of machine translation systems [10]

6.5 Explain Text summarization in detail [10] (Frequency: 3)

6.6 Explain the Information retrieval system [10] (Frequency: 2)

6.7 Explain information retrieval versus Information extraction systems [10]

6.8 Explain Question Answering System with Algorithmic approach [10]

6.9 Explain Question Answering system (QAS) in detail. [10]

6.10 Explain the different steps in text processing for Information Retrieval [10]

Unclassified Questions

(These questions are either too general, combine multiple modules, or pertain to self-learning topics not explicitly detailed in the core syllabus.)

UC.1 Write Short Note [20] a Explain Semi-supervised method (Yarowsky) Unsupervised (Hyperlex) [10] b Explain Question Answering System with Algorithmic approach [10]

(Note: The sub-questions are classified under Modules 4 and 6, but the “Write Short Note” instruction is a meta-question format.)

UC.2 For a given grammar using CYK or CKY algorithm parse the statement [10] “The man read this book” Rules: …

(Note: This is a duplicate of 3.8. It is listed here once as the instruction itself is the question.)

UC.3 Foe given above corpus, S indicates start of the statement and E indicates end of the statement ... Create Transition Matrix & Emission Probability Matrix ... Apply Hidden Markov Model and do POS tagging for given statements

(Note: This is a duplicate of the table-based variant in 3.13. It is listed here once as the instruction itself is the question.)

UC.4 Explain following Syntactic and Semantic Constraints on Co reference 1) Number Agreement 2) Person & Case Agreement

(Note: While related to discourse, its primary focus is on syntactic and semantic constraints, making its module placement (1 vs. 5) ambiguous. It was placed in Module 1.)

UC.5 Explain the significance of regular expression in NLP.

(Note: This is a fundamental tool used across many modules, especially Module 2. It was placed in Module 2.)

This document now includes the marks for each question, allowing you to prioritize your study efforts based on the weightage of each topic.